

Lesson 6

Taxonomy of Data Visualization Methods

Data visualization methods can be categorized based on their primary communication purpose. While there are many purposes, this lesson focuses on two key areas: comparing categories and assessing hierarchies. Other purposes include showing changes over time, plotting connections and relationships, and mapping geo-spatial data.

I. Comparing Categories

This group of visualizations is designed for making comparisons between the relative and absolute sizes of categorical values.

Gantt Chart (Floating Bar)

- **Purpose:** This chart helps to show the range of quantitative values for a given category.
- **Visual Representation:** It uses a horizontal bar that stretches from the lowest value to the highest value for each category. This is effective for comparing the span or duration of different items.

Sankey Diagram

- **Purpose:** Sankey diagrams are primarily used to convey the idea of flow.
- **Visual Representation:** They portray constituent quantities as they move across a number of "stages". The ongoing associations between stages are represented by connecting bands.
- **How it Works:** The width of the connecting links is directly proportional to the quantity of the flow from one stage to another.
- **Use Case:** They are especially useful for visualizing situations where elements transform, divide, or combine over key events.

Small Multiples

- **Purpose:** This technique facilitates efficient and effective comparisons by using a multi-panel display of small, individual chart elements.
- **How it Works:** Small multiples leverage our ability to rapidly scan across a grid (or "trellis") of similar charts to spot patterns and anomalies.
- **Use Case:** They are very effective for comparing categories that have a broad range of values and for showing snapshots of events that change over time.

II. Assessing Hierarchies and Part-to-Whole Relationships

This category of visualizations provides a breakdown of categorical values, showing their relationship to a larger population or as constituent elements of a hierarchical structure.

Circle Packing Diagram

- **Purpose:** This method shows part-to-whole relationships by packing constituent circles (the parts) into an overall circular area (the whole).
- **Visual Representation:** Each individual circle represents a different category. The size of each circle is proportional to its associated quantitative value.
- **Enhancements:** Other visual variables, such as color and position, can be used to add further layers of meaning to the display.

Bubble Hierarchy

- **Purpose:** A bubble hierarchy is specifically used to portray organization and structure through a hierarchical display.
- **Visual Representation:** It uses circles to represent the constituent parts of the hierarchy, such as different departments.
- **How it Works:** The circles are sized according to their quantitative value (e.g., budget, staff count) and are colored to visually distinguish the different departments or categories.